Prospective SUM-159 WGD pilot observables

Specified on 23 October 2026, before SUM-159 WGD+/− motility data; no experimental observations were used.

Model drift subset: f ≤ 0.01 and multiple-domain fraction ≤ 0.05. In cells, classify by the observed frequency of coexisting protrusions; model f and native times have no calibrated cellular equivalents.

Cellular observable	Model counterpart	Drift regime: WGD+ / WGD−	Secondary-domain regime: WGD+ / WGD−	Interpretation
Lateral-edge fluctuation of the leading protrusive region	Edge displacement and drift diffusivity (Figs 3D, 4D)	Similar (Holmes/Spring); smaller in Wave-pinning/Goryachev	No common direction identified outside persistent single-domain bouts	Conditional hypothesis
Secondary-protrusion birth rate per unit perimeter	Persistent no-overlap birth rate (Fig. 4E(i))	Rare in both groups	Model dependent; all-birth size trend unresolved	Population unresolved
Fraction of secondary protrusions that take over	Birth-to-dominance probability by distance (Fig. 4E(ii))	Not applicable when births are too rare	Lower in all four certified subsets within 20τ; all-birth population unresolved	Conditional model result
Activity/protrusion-heading autocorrelation time	Heading decay and size exponent (Fig. 4B–D)	Longer; approximately size squared only when physical drift is size independent	Size advantage shrinks in the tested models; compare coherence and domain counts	Conditional hypothesis
Edge displacement after boundary-straddling photoactivation	Weak deterministic edge response (Fig. 3C)	Smaller (Wave-pinning/Goryachev); similar (Holmes/Spring)	Not specified: the pulse assay uses a stable polarized incumbent	Mechanism probe

These are prospective pilot observables and conditional directions, not observed SUM-159 results or a complete statistical preregistration. Sampling, segmentation, physical pulse calibration and group-level inference must be fixed before comparing the experimental groups.

Birth rates are model dependent: at f = 0.1, identification bounds show an increase with size in Spring and a decrease in Goryachev. Overlapping bounds leave other all-birth trends unresolved. Lower takeover probabilities occur in all four certified subsets at common 20τ follow-up, pooling f > 0.01.

Activity heading, observed cell migration and OU-fitted velocity memory are distinct readouts. Similar physical edge drift implies a size-squared heading advantage only under the single-domain, coherent-heading and negligible-jump assumptions.